

PE-EC703C — Wavelet Transforms

Semester VII • 3L:0T:0P • 3 Credits

Module 1 Exam Notes: Introduction & Continuous Wavelet Transform (CWT)

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1. Introduction and Historical Development

Definition of a Wavelet

A **wavelet** is a localized, oscillatory wave-like function of effectively finite duration, with a zero mean. Unlike a sine wave, which extends infinitely in both directions, a wavelet grows and decays over a short period, providing simultaneous **time** and **frequency** localization.

1.1 Historical Origin of Wavelets

The development of wavelet theory is a synthesis of ideas spanning mathematics, physics, and digital signal processing:

- **1910 (Alfréd Haar):** Proposed the first and simplest wavelet, the **Haar wavelet**, as an orthonormal basis for functions. It consists of a step function that changes value abruptly between $+1$ and -1 .
- **1946 (Dennis Gabor):** Developed the Windowed Fourier Transform (or Gabor Transform) to introduce time localization using Gaussian window envelopes.
- **1981 (Jean Morlet):** A geophysicist processing seismic data introduced the concept of the “wavelet of constant shape” with variable scaling and translation to achieve depth resolution.
- **1984 (Alex Grossmann):** Collaborated with Morlet to establish the rigorous mathematical framework for the Continuous Wavelet Transform (CWT) using group theory.
- **1986 (Stéphane Mallat & Yves Meyer):** Formulated **Multi-resolution Analysis (MRA)**, establishing a unified foundation connecting wavelets to subband digital filtering.
- **1988 (Ingrid Daubechies):** Constructed the first family of continuous, compactly supported, orthonormal wavelet bases with maximum vanishing moments (the **Daubechies wavelets**), making wavelets highly practical for computer implementation.

1.2 Different Communities of Wavelet

Wavelet theory is uniquely interdisciplinary, developed independently across multiple scientific domains:

1. **Mathematics Community:** Studied under *Calderón-Zygmund operators*, singular integrals, and harmonic analysis. Focuses on basis representations of Hilbert and Banach function spaces ($L^2(\mathbb{R})$).
2. **Geophysics & Physics Community:** Morlet and Grossmann used wavelets for seismic reflection parsing and quantum mechanics (coherent states, path integrals).
3. **Digital Signal Processing (DSP) & Engineering Community:** Developed wavelets as **filter banks** (subband coding) for audio, image, and video compression.

2. Time-Frequency Analysis Limitations

Traditional signal analysis relies on the **Fourier Transform (FT)**, which maps a time-domain signal $x(t)$ to its frequency spectrum $X(f)$:

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt$$

- **Limitation:** FT has *infinite time support* ($e^{-j2\pi ft}$ spans from $-\infty$ to ∞). It yields perfect frequency resolution but completely destroys all time information (we know *what* frequencies exist, but not *when* they occur).

2.1 Short-Time Fourier Transform (STFT)

To introduce time localization, the **Short-Time Fourier Transform (STFT)** multiplies the signal with a sliding window function $w(t)$ centered at time τ :

$$STFT_x(\tau, f) = \int_{-\infty}^{\infty} x(t)w^*(t - \tau)e^{-j2\pi ft} dt$$

- **Limitation:** The window width is **fixed** for all frequencies.
- **Heisenberg's Uncertainty Principle:** The resolutions in time (Δt) and frequency (Δf) are bounded:

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$$

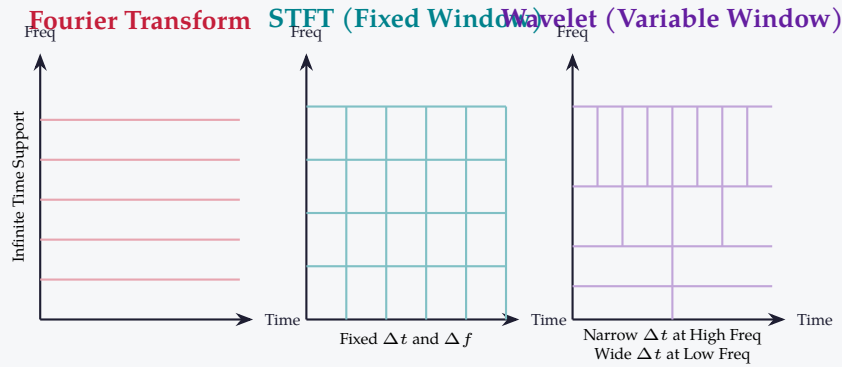
Under STFT, we must choose a fixed window. A narrow window gives good time resolution but poor frequency resolution (Wideband STFT). A wide window gives good frequency resolution but poor time resolution (Narrowband STFT).

2.2 The Wavelet Multi-resolution Approach

The Continuous Wavelet Transform (CWT) resolves this restriction by using a **variable-width window**:

- At **high frequencies**, it uses narrow windows (high time resolution, lower frequency resolution) to capture rapid transients and edges.
- At **low frequencies**, it uses wide windows (high frequency resolution, lower time resolution) to capture slow-varying stationary components.

Time-Frequency Tiling Grids Comparison



3. Continuous Wavelet Transform (CWT)

Definition of CWT

The **Continuous Wavelet Transform (CWT)** of a continuous-time signal $x(t) \in L^2(\mathbb{R})$ is the correlation of the signal with a scaled and translated version of a mother wavelet function $\psi(t)$:

$$CWT_x(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a} \right) dt$$

where:

- $a \in \mathbb{R} \setminus \{0\}$ is the **scale parameter** (controls compression/dilation).
- $b \in \mathbb{R}$ is the **translation parameter** (controls shift along the time axis).
- $\psi(t)$ is the **mother wavelet**, and $\psi^*(t)$ denotes its complex conjugate.
- $\frac{1}{\sqrt{|a|}}$ is the normalization factor ensuring energy is invariant across scales.

3.1 Scaling and Translation Parameters

- **Scaling (a):** Scaling either dilates (stretches) or compresses the mother wavelet:
 - **Large scale** ($|a| > 1$): Wavelet is dilated, representing a low-frequency, long-duration window (captures coarse features).
 - **Small scale** ($|a| < 1$): Wavelet is compressed, representing a high-frequency, short-duration window (captures fine details and transients).
- **Translation (b):** Moves the wavelet along the time axis. It indicates the time center of the window.

3.2 The Constant Q-Factor Filtering Interpretation

The ratio of the center frequency f_c of a filter to its bandwidth Δf is defined as the **Quality Factor (Q-factor)**:

$$Q = \frac{f_c}{\Delta f}$$

In CWT, compressing the wavelet in time by a stretches its Fourier spectrum by $1/a$. Thus, center frequency and bandwidth scale proportionally:

$$f_c(a) = \frac{f_c(1)}{a}, \quad \Delta f(a) = \frac{\Delta f(1)}{a} \implies Q = \frac{f_c(a)}{\Delta f(a)} = \text{Constant}$$

Hence, the CWT behaves as a **constant-Q bandpass filter bank**.

Wavelet Dilation and Compression

Mother Wavelet ($a = 1, b = 0$) Dilated Wavelet ($a = 2.0, b = 0$) Compressed & Shifted ($a = 0.5, b = 3.0$)



4. Mathematical Properties of CWT

4.1 CWT as a Linear Operator

The CWT is a **linear operator**. If $x(t) = c_1x_1(t) + c_2x_2(t)$, then:

$$CWT_x(a, b) = c_1CWT_{x_1}(a, b) + c_2CWT_{x_2}(a, b)$$

4.2 Admissibility Condition

For a function $\psi(t)$ to be a valid mother wavelet, it must satisfy the **admissibility condition**:

$$C_\psi = \int_{-\infty}^{\infty} \frac{|\Psi(f)|^2}{|f|} df < \infty$$

where $\Psi(f)$ is the Fourier transform of $\psi(t)$.

- **Consequence:** For the integral to be finite, $\Psi(0) = 0$. This implies:

$$\int_{-\infty}^{\infty} \psi(t) dt = 0$$

The average value (or DC component) of a wavelet must be **zero**. It must oscillate above and below the zero axis.

4.3 Inverse Continuous Wavelet Transform

If the admissibility condition $C_\psi < \infty$ is met, the signal $x(t)$ can be reconstructed from its wavelet coefficients $CWT_x(a, b)$ via the inverse CWT formula:

$$x(t) = \frac{1}{C_\psi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} CWT_x(a, b) \frac{1}{\sqrt{|a|}} \psi\left(\frac{t-b}{a}\right) \frac{dadb}{a^2}$$

4.4 Energy Conservation (Parseval's Relation)

The wavelet transform preserves energy. If $x(t) \in L^2(\mathbb{R})$, then:

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{C_\psi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |CWT_x(a, b)|^2 \frac{dadb}{a^2}$$

Quick Revision — Module 1

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Key Wavelet Formulas and Metrics

Continuous Wavelet Transform	$CWT_x(a, b) = \frac{1}{\sqrt{ a }} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a} \right) dt$
Admissibility Condition	$C_\psi = \int_{-\infty}^{\infty} \frac{ \Psi(f) ^2}{ f } df < \infty \quad \Rightarrow \quad \int_{-\infty}^{\infty} \psi(t) dt = 0$
Inverse Wavelet Transform	$x(t) = \frac{1}{C_\psi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} CWT_x(a, b) \frac{1}{\sqrt{ a }} \psi \left(\frac{t-b}{a} \right) \frac{da db}{a^2}$
Heisenberg Uncertainty	$\Delta t \cdot \Delta f \geq \frac{1}{4\pi} \quad (\text{Resolution limit in Time-Frequency plane})$
Constant Q-Factor	$Q = \frac{f_c}{\Delta f} = \text{Constant} \quad (\text{Proportional frequency/bandwidth scaling})$

One-Line Definitions

- **Wavelet:** A localized, oscillatory function of finite duration with a zero mean value.
- **CWT:** The correlation of a continuous-time signal with a continuously scaled and translated mother wavelet.
- **Scale Parameter (a):** Controls the compression (small a , high frequency) or dilation (large a , low frequency) of a wavelet.
- **Translation Parameter (b):** Determines the temporal location of the wavelet window center along the time axis.
- **Admissibility Condition:** A mathematical constraint requiring a wavelet's DC component to be zero for reconstruction to be possible.
- **Fourier Transform:** Spectral mapping with infinite time support, providing no temporal location details.
- **STFT:** Time-frequency mapping that uses a fixed window width, imposing rigid time-frequency resolutions.

Frequently Asked / Viva Questions

Q1. Why can the Fourier Transform not analyze non-stationary signals?

The Fourier Transform integrates over all time ($-\infty$ to ∞), meaning it assumes the signal's spectral components do not change over time. It has no time localization, so it cannot identify *when* a transient event or change in frequency occurs.

Q2. How does the Short-Time Fourier Transform (STFT) attempt to fix the limitations of the Fourier Transform?

Frequently Asked / Viva Questions (contd.)

The STFT divides the signal into short segments by multiplying it with a sliding window function. The Fourier Transform is then computed for each segment, providing a time-frequency spectrogram.

Q3. What is the drawback of the STFT?

The STFT uses a fixed window size for all frequencies. Due to Heisenberg's Uncertainty Principle ($\Delta t \cdot \Delta f \geq 1/4\pi$), a narrow window gives good time resolution but poor frequency resolution, while a wide window does the opposite.

Q4. How do Wavelets resolve the resolution trade-off of the STFT?

Wavelets use variable-width windows: narrow windows (small scales) at high frequencies for capturing rapid transitions, and wide windows (large scales) at low frequencies for capturing steady long-term behaviors.

Q5. State the physical meaning of the scale parameter 'a'.

Scale 'a' is inversely proportional to frequency. A small scale ($a < 1$) compresses the wavelet, corresponding to high-frequency transients. A large scale ($a > 1$) dilates the wavelet, corresponding to slow-varying low-frequency components.

Q6. What is the significance of the admissibility condition?

The admissibility condition ensures that the wavelet does not carry a DC component ($\int_{-\infty}^{\infty} \psi(t)dt = 0$). This guarantees that the inverse wavelet transform exists and enables perfect reconstruction of the signal.

Q7. Explain the constant Q-factor filtering property of the CWT.

In CWT, as scale a changes, the center frequency and bandwidth of the wavelet filter change proportionally. Since $Q = f_c/\Delta f$, the quality factor remains constant across all scales, behaving as a logarithmic bandpass filter bank.

Q8. What is the difference between Gabor transform and Wavelet transform?

The Gabor transform is a special case of STFT using a Gaussian window. It uses a fixed window envelope across all frequencies. The Wavelet transform uses a scaled envelope that compresses at higher frequencies and expands at lower frequencies.

Q9. Is the wavelet transform a redundant representation? Why?

Yes. In the Continuous Wavelet Transform, the parameters 'a' and 'b' vary continuously, mapping a 1D time signal $x(t)$ to a 2D time-scale plane $CWT(a, b)$. This generates highly redundant information, which is resolved by discretizing scales and shifts.

Summary Comparison: FT, STFT, and CWT

Feature	Fourier Transform (FT)	Short-Time FT (STFT)	Continuous Wavelet (CWT)
Time Resolution	None (integrates over all time).	Fixed (determined by window width).	Variable (excellent at high frequencies).
Freq Resolution	Perfect (at all frequencies).	Fixed (determined by window width).	Variable (excellent at low frequencies).
Window Shape	Infinite ($e^{-j2\pi ft}$).	Fixed envelope ($w(t)$).	Scaled mother wavelet ($\psi(t/a)$).
Best Use Case	Stationary signals.	Quasi-stationary signals.	Non-stationary/transient signals.